MIL-S-43919C
29 June 1987
SUPERSEDING
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* MILITARY SPECIFICATION

SAUCE AND GRAVY MIXES, DRY, CANNED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers sauce and gravy mixes for use by the Department of Defense as a component of operational rations.

* 1.2 Classification. The product shall be of the following types and classes, as specified (see 6.1).

Type I	- Gravy mixes (commercial)
Class 1	- Mushroom
Class 2	- Chicken
Class 3	- Brown
Type II	- Sauce and seasoning mixes (commercial)
Class 1	- Cheese
Class 2	- Tomato
Class 3	- White
Class 4	- Barbecue
Class 5	- Taco seasoning

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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Type III	- Gravy mixes (extended shelf life)
Class 1	- Mushroom
Class 2	- Chicken flavored
Class 3	- Brown
Type IV	- Sauce and seasoning mixes (extended shelf life)
Class 1	- Cheese
Class 2	- Tomato sauce basic
Class 3	- White
Class 4	- Barbecue
Class 5	- Taco seasoning

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-96	- Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-D-43266	- Desiccants and Desiccation, Method of: For Packaging Subsistence

STANDARDS

FEDERAL

FED-STD-595	- Colors

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

Definitions and Standards for Cereal Flours and Related Products

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

United States Standards for Condition of Food Containers

(Application for copies should be addressed to Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Bulk American Cheese for Manufacturing

U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

U.S. Standards for Grades of Dry Buttermilk

U.S. Standards for Dry Whey

(Application for copies should be addressed to Dairy Division, Room 2750-S, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN OIL CHEMISTS SOCIETY

Official Methods and Recommended Practices of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, P.O. Box 5037, Station A, Champaign, IL 61820.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Rd., St. Paul, MN 55121.)

MUNSELL COLOR CO. INC.

Munsell Color Discs

(Munsell color discs may be obtained from the Munsell Color Co., 2441 No. Calvert, Baltimore, MD 21218.)

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NATIONAL CHEESE INSTITUTE

Cheese Color Reference Standards

(Application for copies should be addressed to the National Cheese Institute, 110 North Franklin Street, Chicago, IL 60606.)

AMERICAN ELSEVIER PUBLISHING COMPANY, INC.

Spot Tests In Inorganic Analysis - by F. Feigl

(Application for copies should be addressed to the Elsevier Publishing Co., Inc., 52 Vanderbilt Ave., New York, NY 10017.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Production

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, P.O. Box 2109, Novato, CA 94947.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 222 South Riverside Plaza, Room 1120, Chicago, IL 60606.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3.2.1.14 Mushrooms, freeze dried. Freeze dried mushrooms shall be processed from fresh, sound, whole, commercially grown, white gill mushrooms (Agaricus campestris).

3.2.1.15 Tomato powder. Tomato powder shall show no evidence of heat damage and shall be free from objectionable odors and extraneous materials. Tomato powder shall be practically free from skins and seeds (may contain skins and seeds in a quantity that no more than slightly affects the appearance of the product). The tomato powder shall be free flowing and of such particle size that 95 percent of the material shall pass through a U.S. Standard No. 20 sieve (0.033 inch openings). Not more than 450 ± 100 parts per million (ppm) of sulfur dioxide shall be allowed in the tomato powder as a preservative.

NOTE: If tomato flakes are used that are of larger granulation than above, they may be mixed with sugar component and ground in a Fitzmill (or similar suitable machine) to reduce the particle size to the required limit. Care must be taken to avoid overheating the agglomeration of the product.

3.2.1.16 Hydrolyzed vegetable protein. The hydrolyzed vegetable protein shall be obtained by the hydrolysis of any one or any combination of corn, soybean, wheat, or yeast proteins from which none of the monosodium glutamate has been extracted. The flavor may be further enhanced with an addition of disodium inosinate and disodium guanylate during processing. It shall have a characteristic odor and flavor. The salt free solids shall be not less than 48.0 percent. The salt content shall be not less than 22.0 and not more than 50.0 percent. Ammonium chloride shall be not more than 1.5 percent. The pH of a three-percent solution shall be not less than 5.0 and not more than 6.3.

3.2.1.17 Chicken meat solids, dried. Dried chicken meat solids shall be a commercial type and shall be produced after the award of contract in a plant operating under supervision of the FSIS. The product shall have good flavor and odor, and shall be reduced to a fine powder to provide proper distribution in the finished product.

* 3.2.1.18 Onion, dehydrated, minced. Dehydrated minced onion shall be derived from white onion varieties having prominent pungent characteristics and shall be free from scorched, musty, and other objectionable odors. It shall be in commercially designated "minced" form. It shall comply with the Official Standards and Methods of ADOGA.

3.2.1.19 Malic acid. Malic acid shall conform to Food Chemicals Codex.

3.2.1.20 Sugar, brown. Brown sugar shall be derived from the refining of cane sugar by controlled evaporation and centrifuging system allowing sugar crystals to retain a portion of the liquors that surround them.

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3.2.1.21 Flavors. Imitation flavors shall be processed in such a manner that the flavors are protected from oxidation, rancidity, evaporation, or other types of deterioration. The flavors may be protected by emulsifying the aromatics of gum acacia mucilage prior to spray drying, or entrapping or encapsulating in vegetable gum, sorbitol, dextrose, or other edible materials.

3.2.1.22 Worcestershire powder. Worcestershire powder shall be a commercially available spray dried product.

* 3.2.1.23 Milk, dry, nonfat. Nonfat dry milk shall be U.S. Extra Grade as defined in the U.S. Standards Grades of Nonfat Dry Milk (Spray Process).

3.2.1.24 Shortening, high solid, beaded type. Shortening shall be high solid beaded shortening made from hydrogenated vegetable oil and shall have an Active Oxygen Method (AOM) stability of not less than 100 hours (see 6.4).

3.2.1.25 Maltodextrins. Maltodextrins shall be a completely soluble product with a dextrose equivalent of not less than 9.0 nor more than 13.0.

3.2.1.26 Dairy ingredients. All dairy ingredients used in the production of these products shall fully comply with the Federal Food, Drug, and Cosmetic Act and applicable Food and Drug Administration's Standard of Identity for that product.

3.2.1.27 Beef extract product. The beef extract product may be imported or may originate in domestic establishments regularly operated under the inspection of the FSIS. The imported product shall be approved by the FSIS. The beef extract product shall have a characteristic flavor and odor and shall not be scorched or burned. The extract shall show no trace of nitrates or nitrites when tested as specified in 4.4.4.

3.2.1.28 Mono and diglycerides. Mono and diglycerides shall be a combination of monoglycerides and diglycerides and shall be a free flowing powder.

* 3.2.1.29 Onion, dehydrated, chopped, toasted. Toasted chopped onions shall be derived from white onions. The product shall be of a size commercially designated as "chopped" and shall have a light brown color and a distinct toasted onion flavor. It shall comply with the Official Standards and Methods of ADOGA.

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g. Type II, class 4 - Barbecue sauce mix
(1) Mandatory ingredients: Starch, dried onion, salt, tomato powder, sugar, and spices.
(2) Optional ingredients: Hydrolyzed vegetable protein, dried garlic, vegetable oil, monosodium glutamate, and flavoring.

h. Type II, class 5 - Taco seasoning mix
(1) Mandatory ingredients: Toasted onion, spices, salt, sugar, monosodium glutamate, and potato starch.
(2) Optional ingredients: Garlic powder, cereal ingredients, cayenne pepper, chili pepper, ground cumin, imitation flavor(s), turmeric, paprika, cocoa, sodium caseinate, and acidulant(s).

3.3.2 Type III and IV (extended shelf life mixes). Type III and IV mixes shall be formulated according to the following formulas:

* 3.3.2.1 Type III - Gravy mixes (extended shelf life).

Ingredients	Class 1 Mushroom gravy mix (percent by weight)	Class 2 Chicken flavored gravy mix (percent by weight)	Class 3 Brown gravy mix (percent by weight)
Wheat flour	16.0 - 18.0	10.0 - 11.0	49.0 - 51.0
Starch	13.0 - 15.0	18.0 - 19.0	8.0 - 12.0
Onion powder 1/	1.5	2.0	1.0
Pepper, white, ground	0.1	0.1	0.05
Buttermilk solids	-	3.0	-
Monosodium glutamate	1.8 - 1.9	1.5	2.0 - 2.5
Maltodextrin	-	7.0 - 8.0	-
Salt mixture	3.4	3.0	2.5
Chicken powder, dehydrated	-	6.5 - 7.5	-
Malic acid	-	0.1	-
Nonfat dry milk solids	30.0 - 34.0	19.0 - 21.0	-
Chicken fat	-	18.0 - 20.0	-
Chicken flavor, imitation	-	as required	-
FD&C color yellow #5	-	0.007	-

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3. REQUIREMENTS

* 3.1 Bid sample approval for types III and IV. A sample consisting of two duplicate No. 2 1/2 (401 x 411) hermetically sealed cans, individually packaged, including sufficient desiccant bags and representative of the product that the contractor proposes to furnish, shall be submitted to the contracting officer prior to bid opening. Each bid sample shall be labelled with type, class, and flavor, as applicable, and net weight. Testing and acceptance of bid samples will include panel testing for palatability as specified (see 6.3). The approved bid sample duplicate shall be used as a standard for palatability of the delivered product. Approved of any bid sample for palatability will not constitute its approval as meeting other requirements of this document.

3.2 Ingredient requirements. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Ingredients for types III and IV product.

3.2.1.1 Flour. Flour shall comply with the appropriate sections of the Requirements for Specific Standardized Cereal Flours and Related Products of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

* 3.2.1.2 Coloring materials. Coloring materials shall be natural, FD&C Certified, Generally Regarded As Safe (GRAS) or a blend thereof.

3.2.1.3 Sugar. Sugar shall be white, refined, granulated cane or beet sugar or a combination thereof.

* 3.2.1.4 Salt or salt mixture. Salt shall comply with Food Chemicals Codex. If salt mixture is used, it shall contain not less than 49.5 percent sodium chloride and not less than 49.6 percent potassium chloride, and shall be free flowing.

3.2.1.5 Spices, ground. Ground spices shall be full strength and shall be typical in appearance, flavor, odor, and pungency of the spice specified, as applicable.

* 3.2.1.6 Dehydrated American cheese. Dehydrated American cheese shall be uniform in composition and appearance, free from lumps that do not fall apart under light finger pressure, and free from burnt specks and foreign material. The hydrated product shall be free from rancid, stale, oxidized, tallowy, bitter, fishy, soapy, scorched, or other objectionable tastes and odors. The

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American cheese used to produce this product shall be U.S. Extra Grade or U.S. Grade A or better of U.S. Standards for Grades of Bulk American Cheese for Manufacturing. The final product shall contain not more than 3.0 percent moisture 5.0 percent salt. The milk fat shall be not less than 48.0 percent. The dehydrated American cheese shall be either colored or uncolored, as required.

* 3.2.1.7 Garlic, dehydrated, powdered. Dehydrated garlic powder shall have a characteristic pungent garlic flavor and odor. It shall be free from scorched, musty, or other objectionable odors. The dehydrated garlic powder shall comply with the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products. Alternatively, when garlic oil is used, it shall be spray dried natural garlic oil, encapsulated with gum arabic, and diluted with dextrose carrier. In addition, oleoresins may be used and shall comply with the Food Chemicals Codex.

* 3.2.1.8 Onion, dehydrated, powdered. Dehydrated onion powder shall be uniformly ground and shall be derived from white varieties having prominent pungent characteristics and shall be free from scorched, musty, and other objectionable odors. The dehydrated onion powder shall comply with the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products. Alternatively, when onion oil is used, it shall be spray dried natural onion oil, encapsulated with gum arabic, and diluted with dextrose carrier. In addition, oleoresins may be used and shall comply with the Food Chemicals Codex.

* 3.2.1.9 Monosodium glutamate. Monosodium glutamate shall be white free-flowing powder and shall comply with the Food Chemicals Codex.

3.2.1.10 Citric acid. Citric acid shall conform to Food Chemicals Codex.

3.2.1.11 Sodium diacetate. Sodium diacetate shall conform to Food Chemicals Codex.

3.2.1.12 Starch. Starch shall be modified waxy maize or modified tapioca starch or a combination thereof.

3.2.1.13 Chicken fat. Chicken fat shall be the product obtained by rendering the wholesome chicken fat under the supervision of the Food Safety and Inspection Service (FSIS). The rendered fat may be protected by the types and amounts of antioxidants approved by the FSIS.

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specification to the extent specified herein. Unless a specific issue

3.2.1.30 Caramel coloring. Caramel coloring shall be dry, readily soluble brown powder obtained from sucrose or glucose.

3.2.1.31 Potato starch. Potato starch shall be uniformly white, finely pulverized powder derived from sound, white potatoes.

* 3.2.1.32 Disodium inosinate and disodium guanylate. Disodium inosinate and disodium guanylate shall be free flowing powder produced by enzymatic treatment of yeast ribonucleic acid and shall comply with the Food Chemicals Codex.

3.2.1.33 Smoke flavor, natural. Natural smoke flavor shall be approved by the FSIS. The natural smoke flavor shall be liquid or powder and impart a pleasing smoke flavor when used at the formula level designed for the product.

* 3.2.1.34 Whey, dry. Dry whey shall be U.S. Extra Grade Dry Sweet Type Whey as defined in the United States Standards for Dry Whey.

* 3.2.1.35 Buttermilk, sweetcream, dry. Dry sweetcream buttermilk shall be U.S. Extra Grade as defined in the U.S. Standards for Grades of Dry Buttermilk.

* 3.2.1.36 Anticaking agent. Anticaking agent shall be silicon dioxide or its equivalent, and shall comply with the Food Chemicals Codex. It shall be the low pH product.

3.3 Formulation.

3.3.1 Types I and II (commercial mixes). Types I and II shall be dry mixes consisting of the following materials in any proportion or order of ingredient and shall be so handled and mixed that the final product shall be a complete homogeneous mix.

a. Type I, class 1 - Mushroom gravy mix
(1) Mandatory ingredients: Hydrolyzed vegetable protein, starch, dried mushrooms, salt, spices, caramel color, and extractive of mushrooms.
(2) Optional ingredients: Cereal ingredients, tapioca flour, hardened vegetable oil, beef fat, sugar, corn syrup solids, monosodium glutamate, dried onion, nonfat dry milk solids, buttermilk solids, phosphate, artificial flavors, and preservatives.

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b. Type I, class 2 - Chicken gravy mix
(1) Mandatory ingredients: Starch, monosodium glutamate, and chicken fat.
(2) Optional ingredients: Cereal ingredients, hydrolyzed vegetable protein, buttermilk solids, nonfat dry milk solids, cream solids, dried onion, salt, lactose, dextrose, sugar, yeast, phosphates, preservative, spices, artificial color and flavor, and dried chicken broth or dried chicken meat or a combination thereof.

c. Type I, class 3 - Brown gravy mix
(1) Mandatory ingredients: Starch, hydrolyzed vegetable protein, and salt.
(2) Optional ingredients: Tapioca flour, cereal ingredients, beef extract, sugar, monosodium glutamate, shortening, hardened vegetable oil, beef fat, dried cream, whey solids, maltodextrins, sodium caseinate, buttermilk solids, nonfat dry milk solids, dried onions, beet powder, caramel color, spices, artificial flavor, garlic, phosphates, and preservative.

d. Type II, class 1 - Cheese sauce mix
(1) Mandatory ingredients: Cheese, starch, and salt.
(2) Optional ingredients: Nonfat dry milk solids, whey solids, hydrolyzed vegetable protein, monosodium glutamate, dried onion, mustard, garlic, spices, phosphates, lactic acid, stabilizer, artificial color, and flavor.

e. Type II, class 2 - Tomato sauce mix
(1) Mandatory ingredients: Tomato powder, spices, salt, sugar, starch, and garlic.
(2) Optional ingredients: Dried onion, hydrolyzed vegetable protein, monosodium glutamate, cereal ingredients, nonfat dry milk solids, herbs, milk, sugar, dried yeast, cheese, dried mushrooms, sweet pepper, artificial flavor, beet powder, and stabilizer.

f. Type II, class 3 - White sauce mix
(1) Mandatory ingredients: Cereal ingredients, nonfat dry milk solids, starch, and salt.
(2) Optional ingredients: Hardened vegetable oil, hydrogenated vegetable oil, emulsifiers, stabilizers, citrate, buttermilk solids, hydrolyzed vegetable protein, onion, spices, yeast, chicken fat, artificial flavor, and seasonings.

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Ingredients (continued)	Class 1 Mushroom gravy mix (percent by weight)	Class 2 Chicken flavored gravy mix (percent by weight)	Class 3 Brown gravy mix (percent by weight)
Celery powder	0.15	-	0.05
Shortening	4.5	-	10.0 - 12.0
Caramel color	as required	-	as required
Mushroom powder	7.0 - 9.0	-	-
Mushroom, diced	4.0 - 6.0	-	-
Mushroom, flavor	as required	-	-
Sodium caseinate	2.0 - 3.0	-	-
Hydrolyzed vegetable protein	4.0 - 4.5	6.5 - 7.5	9.5 - 11.5
Sugar	-	-	4.0 - 5.0
Paprika, ground	-	-	0.4 - 0.6
Beef extract product	-	-	2.0 - 2.5
Disodium inosinate and disodium guanylate	0.02	0.01	0.08 - 0.11
Anticaking agent	2(max)	2(max)	2 (max)

1/ Onion powder replacement shall be spray dried onion oil, encapsulated with gum arabic and diluted with dextrose carrier. Onion powder replacement shall be used at the level of 100 percent substitution.

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Ingredients (continued)	Class 1 Cheese sauce mix (percent by weight)	Class 2 Tomato sauce, basic mix (percent by weight)	Class 3 White sauce mix (percent by weight)	Class 4 Barbecue sauce mix (percent by weight)	Class 5 Taco seasoning mix (percent by weight)
Paprika, ground	-	-	-	-	5.0 - 7.0
Starch, potato	-	-	-	-	4.5 - 5.5
Cayenne pepper, ground	-	-	-	-	1.25
Oregano, ground	-	-	-	-	5.2 - 6.0
Cumin seed, ground	-	-	-	-	5.0 - 6.0
Flavor, tomato, imitation	-	-	-	-	as required
Turmeric, ground	-	-	-	-	as required
Smoke flavor natural	-	-	-	0.25 - 0.5	-
Disodium inosinate and disodium guanylate	0.01	-	0.02	-	0.04 - 0.06
Anticaking agent	2(max)	2(max)	2(max)	2(max)	2(max)

1/ Garlic powder replacement shall be garlic oil, spray dried, encapsulated with gum arabic and diluted with dextrose carrier. Garlic powder replacement shall be used at the level of 100 percent substitution.

2/ Onion powder replacement shall be onion oil, spray dried, encapsulated with gum arabic and diluted with dextrose carrier. Onion powder replacement shall be used at the level of 100 percent substitution.

3.4 Finished product requirements. The finished product shall be a complete homogeneous mix, free of lumps that cannot be broken up by light pressure, and shall comply with requirements specified in 3.4.1 through 3.4.2.3, as applicable to the pertinent types and classes of product.

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3.4.1 Granulation. The finished product shall consist of particulate granules of such size (see 4.4.1) that will comply with the following requirements for percentage granulation by weight:

a. Type III - Gravy mixes:

Class 1 - Minimum of 95 percent shall pass through a U.S. Standard No. 25 sieve.
Class 2 - Minimum of 95 percent shall pass through a U.S. Standard No. 25 sieve.
Class 3 - Minimum of 95 percent shall pass through a U.S. Standard No. 20 sieve.

b. Type IV - Sauce mixes:

Class 1 - Minimum of 95 percent shall pass through a U.S. Standard No. 25 sieve.
Class 3 - Minimum of 95 percent shall pass through a U.S. Standard No. 25 sieve.

3.4.2 Prepared product. When prepared in accordance with 3.4.2.1, gravies and sauces shall have a smooth texture and be practically free of lumps or graininess. Prepared sauces and gravies shall possess no burnt or objectionable flavors or odors. Taco seasoning, when prepared with ground meat in accordance with 3.4.2.1.3 or 3.4.2.1.4, shall be free from any objectionable flavor or odor.

3.4.2.1 Preparation of gravies, sauces, and seasonings.

3.4.2.1.1 Types I and II (except type II, class 5). Types I and II shall be prepared in accordance with directions on the package except that, when applicable, a proportionate quantity of product and liquid shall be utilized to make approximately 8 ounces (1 cup) of cooked product (see 3.4.2.1.3 for type II, class 5).

3.4.2.1.2 Types III and IV (except type IV, class 5). Types III and IV shall be prepared in the proportions as listed below utilizing 2 ounces of gravy or sauce mix. After dilution, the product shall be cooked over medium heat, stirring constantly until starch and flour, as applicable, completely gelatinize (see 3.4.2.1.4 for type IV, class 5).

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* 4.3.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.3.1.2 Component testing. In addition to any testing required by 4.3.1, components shall be tested in accordance with table I. Any test failure shall be cause for rejection of the related component lot. Use of such component shall be cause for rejection of the involved quantity of finished product.

TABLE I. Component tests

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt para 3/
Chicken fat	1 pound composite 1/	Pounds	S-1	Peroxide value	Nearest meq/kg	3.2.1.13 4.4.3
Tomato powder	1 pound composite 1/	Pounds	S-1	SO_2	Nearest ppm	3.2.1.15 4.4.3
	100 gram composite 1/	Pounds	S-1	Particle size	Pass or fail	3.2.1.15 4.4.1

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3.3.2.2 Type IV - Sauce and seasoning mixes (extended shelf life).

Ingredients	Class 1 Cheese sauce mix (percent by weight)	Class 2 Tomato sauce, basic mix (percent by weight)	Class 3 White sauce mix (percent by weight)	Class 4 Barbecue sauce mix (percent by weight)	Class 5 Taco seasoning mix (percent by weight)
Flour, wheat	-	-	14.0 - 15.0	-	-
Tomato powder	-	62.0 - 64.0	-	34.0 - 36.0	-
Cheese, powdered (colored)	25.0 - 27.0	-	-	-	-
Cheese, powdered (uncolored)	25.0 - 27.0	7.0 - 8.0	-	-	-
Sugar, brown	-	-	-	9.0 - 11.0	-
Starch	19.0	7.0 - 8.0	10.5 - 11.5	13.0 - 14.0	-
Salt or salt mixture	3.0 - 3.5	6.0 - 6.5	3.5	7.5 - 8.5	14.0 - 15.0
Maltodextrin	-	4.0	-	-	17.0
Hydrolyzed vegetable protein	-	1.5 - 3.0	-	-	-
Garlic powder 1/	-	0.5	-	0.1	4.0 - 6.0
Red pepper powder	-	0.01	-	0.02	-
Sugar	-	4.5	-	20.0	-
Monosodium glutamate	1.2	-	1.7	-	1.8 - 2.3
Oregano powder	-	0.1	-	-	-
Onion, minced, flaked	-	2.0	-	-	-
Citric acid	-	0.05	-	1.0	1.0
Coloring (dry canthaxanthin beadlets)	as required	as required	-	as required	-
Basil, ground	-	0.1	-	-	-
Thyme, ground	-	0.1	-	-	-
Rosemary, ground	-	0.1	-	-	-

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Ingredients (continued)	Class 1 Cheese sauce mix (percent by weight)	Class 2 Tomato sauce, basic mix (percent by weight)	Class 3 White sauce mix (percent by weight)	Class 4 Barbecue sauce mix (percent by weight)	Class 5 Taco seasoning mix (percent by weight)
Marjoram, ground	-	0.1	-	-	-
Sodium diacetate	-	-	-	1.0 - 2.0	-
Cloves, ground	-	-	-	0.05	-
Onion, dehydrated, powdered 2/	-	-	1.5	5.9	-
Allspice, ground	-	-	-	0.03	-
Pepper, black, ground	-	-	-	0.50	-
Mustard, prepared	0.25	-	-	3.7	-
Vegetable shortening	-	-	16.0 - 18.0	0.5	-
Worcestershire powder	-	-	-	3.5 - 4.5	-
Whey powder	19.0 - 21.0	-	-	-	-
Pepper, white	-	-	0.1	-	-
Nonfat, dry milk solids	-	-	45.0	-	-
Sodium caseinate	-	-	1.0 - 2.0	-	as required
Flavor, butter, imitation	-	-	0.4	-	-
Mono and diglyceride	-	-	1.5	-	-
Dry-APO-carotene (10 percent)	as required	-	0.003	-	-
Onion, toasted, chopped	-	-	-	-	19.0 - 21.0
Chili pepper, California	-	-	-	-	9.0 - 11.0

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ping shall be limited to the type and class of food specified (see 6.2). Strap

Item	Ratio by weight
Type III - Gravy mixes (extended shelf life)	
Class 1 - Mushroom	1:8 ratio of mix to water
Class 2 - Chicken flavored	1:7 ratio of mix to water
Class 3 - Brown	1:9 ratio of mix to water
Type IV - Sauce mixes (extended shelf life)	
Class 1 - Cheese	1:6 ratio of mix to milk
Class 2 - Tomato sauce, basic	1:5 ratio of mix to water
Class 3 - White	1:5 ratio of mix to water
Class 4 - Barbecue	1:4 ratio of mix to water

* 3.4.2.1.3 Type II, class 5. The commercial taco seasoning shall be prepared in accordance with directions on the package except that 1/2 pound of ground beef shall be utilized with a proportionate amount of water and taco seasoning mix.

3.4.2.1.4 Type IV, class 5. The extended shelf life taco seasoning shall be prepared as follows:

a. Brown 1/2 pound ground beef in a skillet, stirring constantly until the meat is crumbly. Drain off excess fat.
b. Stir 1/2 cup of water and 1 ounce of taco seasoning mix into the meat.
c. Bring to a boil and simmer for approximately 10 minutes.

3.4.2.2 Color. The prepared gravies and sauces of the types and classes specified shall have the following color requirements (see 4.4.2):

a. Type III - Gravy:

Class 1 - Not lighter than 20219 nor darker than 20059 of FED-STD-595.
Class 2 - Not lighter than 23695 of FED-STD-595 nor darker than Munsell color disc designated 2.5y 8/4 (yellow).
Class 3 - Not lighter than 20140 nor darker than 20059 of FED-STD-595.

b. Type IV - Sauce:

Class 1 - Not lighter than Munsell color disc designated 7.5 YR 8/8 (moderate orange yellow) nor darker than Munsell color disc designated 5 YR 8/7 (light orange).

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Class 2 - Not lighter than 22203 nor darker than 22144 of FED-STD-595.
Class 3 - Not lighter than 27855 nor darker than 23727 of FED-STD-595.

3.4.2.3 Viscosity (type IV, class 2). When prepared in accordance with 3.4.2.1.2, the basic tomato sauce shall have a consistency such that it shall flow not more than 10.0 nor less than 6.0 centimeters in a Bostwick Consistometer when tested as specified in 4.3.5.

* 3.4.2.4 Palatability. The product shall be equal to or better than the approved bid sample in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the Title 21, Code of Federal Regulations, Part 110: Current Good Manufacturing Practice in Manufacture, Processing, Packaging or Holding of Human Foods; and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

* 4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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TABLE I. Component tests - Continued

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt para 3/
Hydrolyzed vegetable protein	1 pound composite 1/	Pounds	S-1	Ammonium chloride, pH, sodium chloride, salt-free solids (total solids minus salt content)	Nearest 0.1 percent	3.2.1.16 4.4.3
Vegetable shortening	1 pound composite 1/	Pounds	S-1	Stability	Pass or fail 2/	3.2.1.24 4.4.3
Maltodextrins	1 pound composite 1/	Pounds	S-1	Dextrose equivalent	Nearest whole number	3.2.1.25 4.4.3
Beef extract	1 pound composite 1/	Pounds	S-1	Nitrate Nitrite	Pass or fail	3.2.1.27 4.4.4

1/ The composite shall be formed from the number of primary containers indicated by the inspection level.

2/ If failure, report results to nearest hour.

3/ Unless otherwise specified, a certificate of conformance shall be submitted.

4.3.1.3 Can tin coating testing. The cans shall be tested for tin coating weight (specified in 5.1.1) in accordance with PPP-C-96.

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4/ Determination of the defects for types III and IV shall be made with reference to the bid sample (see 3.1).

5/ Determination of the defects for types III and IV shall be made with reference to the bid sample (see 3.1) and color specification (see 3.4.2.2).

4.3.5 Viscosity testing (type IV, class 2 only). The finished product shall be tested for conformance to viscosity requirement in 3.4.2.3. The sample shall be the number of cans indicated by inspection level S-2. A 2-ounce specimen from each can in the sample shall be reconstituted as specified in 3.4.2.1.2 and tested for flow in a Bostwick Consistometer for 30 seconds at 80° ± 5°F. The results shall be reported to the nearest 0.1 centimeter. Any test failure shall be cause for rejection of the lot.

4.3.6 Primary container condition examination. Examination of filled and sealed primary containers shall be in accordance with the United States Standards for Condition of Food Containers.

4.3.7 Can leakage testing. The filled and sealed cans shall be tested for leakage by submerging in water contained in a vacuum desiccator or equivalent device, drawing a vacuum of 10 inches of mercury, holding for 30 seconds, and observing for leakage. Leakage is indicated by a steady progression of bubbles. Isolated or static bubbles caused by entrapped air in seams or on surfaces are not considered leakage. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-3. Any sign of leakage shall be cause for rejection of the lot.

* 4.3.8 Examination of fiberboard shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows:

Major:	National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible
	Reinforcement with nonmetallic strapping or tape is not used
	Tiered cans not separated with pads
Minor:	Other required markings missing, incorrect, or illegible
	Number or arrangement of cans not as specified

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Level C shipping containers shall be examined only for the marking, arrangement, and number of can defects specified above, and for the closure method specified in 5.2.3.

4.3.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

4.4 Methods of inspection.

4.4.1 Sieving procedure. Place a 100-gram sample on the sieve, including cover and catch pan. Hold the sieve assembly with the left hand at an angle of about 30 degrees from the horizontal, so that the edge of the tilted plane is toward the right hand. Shake by striking the right side of the sieve assembly with a short sharp stroke with the right hand moving in the horizontal plane. Strike at the rate of 150 times per minute. After each 25 strokes, turn the sieve assembly 1/6 of the revolution in the same direction. Continue shaking for 3 minutes. Remove, weigh, and calculate the percentage of sample passing through the sieve.

4.4.2 Color comparison of the prepared gravies and sauces. The prepared gravies and sauces shall be poured into 150 mL beaker and shall be compared with designated Munsell discs and FED-STD-595 chips under daylight of Northern exposure.

* 4.4.3 Chemical analysis test methods. Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists except as otherwise indicated.

Test	Source		Method
Tomato powder			
Sulfur Dioxide	Chapter:	Food Additives: Direct	Total Sulfurous Acid
	Section:	Chemical Preservatives: Sulfurous Acid	
Vegetable shortening			
Fat stability (AOM)	1/		Cd 12-57

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TABLE V. Theoretical moisture content

Mix		Theoretical moisture content (percent by weight)
Type	Class	
III	1	3.5
III	2	4.0
III	3	4.5
IV	1	3.0
IV	2	2.0
IV	3	3.5
IV	4	2.1
IV	5	3.0

5.1.2 Level C.

5.1.2.1 Types I and II product. Types I and II product shall be unit packed in a size 401 by 411 can and preserved in a manner to protect it against damage during shipment from the supply source to the first consuming activity. The contractor may use his standard practice when it meets this requirement.

5.1.2.2 Types III and IV product. Types III and IV product shall be unit packed in accordance with 5.1.1 except that cans with or without commercial exterior coating are acceptable. Alternatively, the cans may be made from 0.20 pound per base box electrolytic tinplate, provided the can is coated with a commercial exterior coating.

5.2 Packing. Product shall be packed in accordance with level A, B, or C requirements, as specified (see 6.1 and 6.2). Twenty-four size 401 by 411 cans arranged four in length, three in width, and two in depth, shall be packed on end in a snug-fitting container.

* 5.2.1 Level A packing. The shipping container shall be a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, V2s, of PPP-B-636. The tiered cans shall be separated with a full length and width pad. Each shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in

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* 4.3.2 In-process examination. When specified (see 6.1), in-process examination shall be performed to determine conformance to the formulation and desiccant requirements. Any nonconformance revealed by actual examination or by review of records of formulation and calculation of desiccant amounts shall be cause for rejection of the involved product.

4.3.3 Net weight inspection. The filled primary containers shall be inspected for net weight in accordance with table II. Each type and class of product shall be examined as a separate lot. The lot size shall be expressed in units of primary containers. The sample unit shall be one filled primary container. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

TABLE II. Net weight defects 1/

Category	Defect
Minor	
201	More than 5 percent under specified net weight (containers required to have 1 pound or less)
202	More than 2 percent under specified net weight (containers required to have more than 1 pound)

1/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight. The net weight shall be reported to nearest 0.1 ounce.

* 4.3.4 Product and desiccant inspection. The contents of the filled primary containers shall be inspected for the defects listed in table III. Each type and class of product shall be examined as a separate lot. The lot size shall be expressed in units of primary containers. The sample unit shall be one filled primary container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

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TABLE III. Finished product and desiccant defects 1/

Category	Defect
Major	Desiccant bag (types III and IV product)
101	Desiccant bag(s) missing
102	Desiccant bag(s) has tear, hole, or open seam
103	Desiccant bag(s) folded or crimped
	Dry mix
104	Not a uniform mixture
105	Presence of lumps that do not break up with light finger pressure
106	Granulation not as specified in 3.4.1 (type III, all classes, and type IV, classes 1 and 3) 2/
	Cooked product 3/
107	Texture lumpy or grainy (types I, II, III and IV) (except type II, class 5, and type IV, class 5) 4/
108	Color not as required (type III, classes 1, 2 and 3, and type IV, classes 1, 2 and 3) 5/

1/ Presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), or undesirable off-flavor or odor or foreign odor shall be cause for rejection of the lot.

2/ See 4.4.1 for procedure. Make one determination per sample unit.

3/ See 3.4.2.1 for cooking procedure. Make one determination per sample unit.

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Test	Source		Method
Hydrolyzed vegetable protein			
pH	Chapter:	Beverages: Malt Beverages and Brewing Materials	Hydrogen-Ion Activity (pH)
	Section:	Beer	
Salt	Chapter:	Meat and Meat Products	Chloride
	Section:	Meat Extracts and Similar Products	
Moisture	Chapter:	Cereal Food	Vacuum Oven
	Section:	Wheat Flour	
Ammonium chloride	Chapter:	Fertilizer	Ammoniacal Nitrogen (Magnesium Oxide)
	Section:	Nitrogen	
Total salt free solids	Chapter:	Sugars and Sugar Products	Vacuum Drying
	Section:	Sugars and Sirup	
Maltodextrin	Chapter:	Sugars and Sugar Products	Total Reducing Sugar
Total reducing sugar (dextrose equivalent) 2/	Section:	Corn Sirups and Sugars	

1/ Official Methods of the American Oil Chemists' Society.

2/ Dextrose equivalents shall be the total reducing sugars expressed as percent of total dry substance.

4.4.4 Nitrates and nitrites. Qualitative tests for nitrates shall be made as specified in test 1, of Spot Test in Inorganic Analysis (F. Feigl), 1954 edition. Qualitative tests for nitrites shall be made as specified in test 1.

5. PACKAGING

5.1 Preservation. Types I and II product shall be preserved in accordance with level C. Types III and IV product shall be preserved in accordance with level A or C, as specified (see 6.1).

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5.1.1 Level A. The type and class of product specified (see 6.1), in quantity and can size specified in table IV, shall be filled into an open top style, round, metal can with soldered or welded side seam, and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25 pound electrolytic tin plate per base box (see 4.3.1.3) and shall be coated overall on the outside with a coating complying with type I, or when specified (see 6.1) type III of TT-C-495. The can shall have a sufficient base plate weight and temper to protect the product during shipment and storage. Each can shall be provided with sufficient desiccant complying with MIL-D-43266 to reduce the moisture content to the theoretical level specified in table V. The amount of desiccant required to achieve the theoretical moisture content shall be calculated from the formula specified in MIL-D-43266, except that the value for percent water as used in the formula shall be the actual moisture content of the product as determined in accordance with The Vacuum Oven Method in the Official Methods of Analysis of the Association of Official Analytical Chemists, minus the theoretical moisture content specified in table V. A record of the calculations shall be maintained. The can shall not leak when examined in accordance with 4.3.7.

TABLE IV. Net weight and can size for extended shelf life mixes

Product	Net wt. 401 by 411 can
Mushroom gravy	11 oz
Chicken flavored gravy	1 lb 2 oz
Brown gravy	1 lb 3 oz
Cheese sauce	1 lb
Tomato sauce, basic	1 lb
White sauce	1 lb
Barbecue sauce	1 lb
Taco seasoning	1 lb 5 oz

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accordance with MIL-L-35078 for the type and class of load specified (see 6.1). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. The shipping container shall be a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, V3c, V3s, or V4s of PPP-B-636. The tiered cans shall be separated with a full length and width pad. Each shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except the closure of the fiberboard boxes shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.3 Unit loads. When specified (see 6.2), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping except for type II, class F loads.

5.4 Labeling and marking.

5.4.1 Types I and II product. Labeling of types I and II product shall be in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder, and shall include directions for use.

5.4.2 Types III and IV product. Types III and IV product in cans shall be labeled in accordance with MIL-L-1497. Labeling shall include directions for use as specified in 5.4.2.1 through 5.4.2.8. Following directions for use for type III and IV products, the words "Remove and Discard Desiccant Bag(s)" shall be included on the label.

DIRECTIONS FOR USE

5.4.2.1 Mushroom gravy mix. Combine contents of can (11 oz) with 3 cups cold water. Mix until smooth. Add to 2 quarts boiling water, stirring constantly. Cook over medium heat until gravy comes to boil. Simmer 1 minute or until thickened. Stir as necessary. Yield: 2 3/4 quarts.

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* 6.5 Subject term (key word) listing.

Barbecue sauce mix
Brown gravy mix
Cheese sauce mix
Chicken gravy mix
Gravy mixes
Mushroom gravy mix
Sauce mixes
Seasoning mixes
Taco seasoning mix
Tomato sauce mix
White sauce mix

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0574

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Civil Agencies Coordinating Activity:

VA - OSS

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5.4.2.2 Chicken gravy mix. Combine contents of can (1 lb 2 oz) with 1 quart cold water. Mix until smooth. Add to 3 quarts boiling water, stirring constantly. Cook over medium heat until gravy comes to a boil. Simmer 1 minute or until thickened, stir as necessary. Yield: 1 gallon.

5.4.2.3 Brown gravy mix. Combine contents of can (1 lb 3 oz) with 1 quart cold water. Mix until smooth. Add to 1 gallon boiling water, stirring constantly. Cook over medium heat until gravy comes to a boil. Simmer 1 minute or until thickened, stir as necessary. Yield: 5 1/4 quarts.

5.4.2.4 Cheese sauce mix. Combine contents of can (1 lb) with 1 quart cold milk. Mix until smooth. Add 2 quarts hot milk, stirring constantly. Cook over medium heat until sauce comes to a simmer. Simmer 1 minute or until thickened, stir as necessary. Yield: 3 quarts.

NOTE: Use 10 ounces (1 1/4 cups) nonfat dry milk, 3 1/2 ounces (7 tbsp) butter or margarine, and 2 3/4 quarts water for milk.

5.4.2.5 Tomato sauce, basic mix. Combine contents of can (1 lb) with 1 quart cold water. Mix until smooth. Add to 1 1/2 quarts boiling water, stirring constantly. Cook over medium heat until sauce comes to boil. Simmer 1 minute or until thickened, stir as necessary. Yield: 2 1/2 quarts.

5.4.2.6 White sauce mix. Combine contents of can (1 lb) with 1 quart of cold water. Mix until smooth. Add to 1 1/2 quarts of boiling water, stirring constantly. Cook over medium heat until sauce comes to a boil. Simmer 1 minute or until thickened, stir as necessary. Yield: 2 1/2 quarts.

5.4.2.7 Barbecue sauce mix. Combine contents of can (1 lb) with 1 quart cold water. Mix until smooth. Add to 1 quart boiling water, stirring constantly. Cook over medium heat until sauce comes to a boil. Simmer 1 minute, stir as necessary. Yield: 2 quarts.

5.4.2.8 Taco seasoning mix. Cook 12 lbs ground beef until it loses its pink color, stirring to break meat in to small pieces. Drain or skim off fat. Add contents of can (1 lb 5 oz) and 2 1/2 quarts of water. Simmer 15 minutes. Yield: 7 1/2 quarts.

5.4.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Purchasers shall exercise any desired options offered herein, and acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and class of product required (see 1.2).
c. When in-process examination is required (see 4.3.2).
d. Levels of preservation and packing required (see 5.1 and 5.2).
e. When type III exterior coating of cans is required (see 5.1.1).
f. Type and class of unit load required (see 5.2.1).
g. Type and class of unitized load when unit loading is specified (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipping, handling, or storage of the specified item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

* 6.3 Bid sample evaluation. Bid samples will be evaluated by U.S. Army Natick, Research, Development, and Engineering Center. The bid sample will be rated by a consumer-type panel on a 9-point hedonic scale graduated in successive degrees of "like extremely" to "dislike extremely". For further details of the evaluation procedure, see "Hedonic Scale Method of Measuring Food Preference" from Food Technology, 1957, Vol. XI, No. 9. Copies may be obtained from the U.S. Army Natick Research, Development, and Engineering Center (ATTN: STRNC-WBF), Natick, MA 01760-5014.

6.4 Vegetable shortening. Shortening with the following solid fat index has been found to be suitable:

Temperature °F	Percent solids
50	80
70	80
80	80
92	80
100	77
110	62

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